

Phase Field Theory of Inequality and Consensus

Allen-Cahn / Cahn-Hilliard Dynamics on SCX Potential Manifolds

不平等与共识的相场理论

SCX 势能流形上的 Allen-Cahn/Cahn-Hilliard 动力学

Phase Separation, Nucleation, Coarsening, and Theorem Connections

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Abstract

We formulate the dynamics of inequality and consensus in the SCX (Situs Consensus eXpert) framework as a **phase field theory**. The attitude/gauge field $\mathbf{g}(x, t)$ and the potential field $S(x, t)$ are treated as coupled order parameters on the Situs manifold $\mathcal{M}$. Drawing from the classical Allen-Cahn and Cahn-Hilliard equations of materials science, we derive:

- (1) **Allen-Cahn dynamics for $\mathbf{g}$:** $\partial_t \mathbf{g} = -\delta \mathcal{F} / \delta \mathbf{g}$ —the gauge field evolves via non-conserved gradient flow to minimize a Ginzburg-Landau free energy, forming honest ($\mathbf{g} = \mathbf{0}$) and dishonest ($\mathbf{g} \neq \mathbf{0}$) domains separated by diffuse interfaces of thickness δ_c ;
- (2) **Cahn-Hilliard dynamics for S :** $\partial_t S = \nabla \cdot (M(S) \nabla (\delta \mathcal{F} / \delta S))$ —the potential field undergoes mass-conserving evolution, redistributing potential while preserving total “social energy,” directly modeling inequality dynamics;
- (3) **Three-phase diagram** in $(\|\mathbf{g}\|, S)$ space: (a) EQUALITY phase — $\|\mathbf{g}\| \approx 0$, smooth S ; (b) STRATIFIED phase — $\|\mathbf{g}\| > 0$, S exhibits plateaus; (c) EXPLOSIVE phase — $\|\mathbf{g}\|$ diverges, S gradients blow up. Phase boundaries are the **inequality phase transition lines**;
- (4) **Nucleation and growth:** A small honest cluster (“audit nucleus,” $\mathbf{g} = \mathbf{0}$ domain) can grow only if its radius exceeds the critical radius $R_c \propto \sigma / \Delta f$, where σ is the interface energy and Δf is the bulk free energy gain. We derive R_c from the competition between surface and volume terms;
- (5) **Coarsening dynamics:** After a social “quench” (rapid institutional change), honest domains coarsen via Ostwald ripening —large honest domains grow at the expense of small ones. The characteristic domain size follows $L(t) \sim t^{1/3}$, the classical Lifshitz-Slyozov-Wagner (LSW) scaling;
- (6) **Theorem connections:** Theorem 11 (attitude singularity) corresponds to a first-order phase transition at the spinodal; Theorem 10 (boundary locking) maps to **pinning of domain walls** at heterogeneities; Theorem 12 (Matthew staircase) maps to **nucleation on defects** —staircase steps act as preferential nucleation sites that accelerate system destabilization.

We provide a complete verification script (`verify_phase_field.py`) that numerically solves the coupled Allen-Cahn / Cahn-Hilliard system on a 2D grid, confirms

the $t^{1/3}$ coarsening scaling, computes the phase diagram, and validates the critical nucleus prediction.

我们将 *SCX* (*Situs* 共识专家) 框架中的不平等与共识动力学表述为**相场理论**。态度/规范场 $\mathbf{g}(x, t)$ 和势能场 $S(x, t)$ 被处理为 *Situs* 流形 $\mathcal{M}$ 上的耦合序参量。借鉴材料科学中经典的 *Allen-Cahn* 和 *Cahn-Hilliard* 方程, 我们推导出: (1) $\mathbf{g}$ 的 **Allen-Cahn 动力学** —规范场通过非守恒梯度流演化以最小化 *Ginzburg-Landau* 自由能, 形成诚实地带 ($\mathbf{g} = \mathbf{0}$) 和不诚实地带 ($\mathbf{g} \neq \mathbf{0}$) 的畴结构; (2) S 的 **Cahn-Hilliard 动力学** —势能场经历质量守恒的演化, 在保持总”社会能量”的同时重新分配势能; (3) 在 $(\|\mathbf{g}\|, S)$ 空间中的**三相图** —平等相、分层相和爆炸相; (4) **成核与生长** —审计核的临界半径; (5) **粗化动力学** — $L(t) \sim t^{1/3}$ 的 *LSW* 标度律; (6) **定理联系** —*Thm11* 对应相变、*Thm10* 对应畴壁钉扎、*Thm12* 对应缺陷成核。

Keywords: phase field theory, Allen-Cahn equation, Cahn-Hilliard equation, inequality dynamics, consensus, nucleation, coarsening, Ostwald ripening, spinodal decomposition, SCX framework

关键词: 相场理论, Allen-Cahn 方程, Cahn-Hilliard 方程, 不平等动力学, 共识, 成核, 粗化, Ostwald 熟化, 旋节分解, SCX 框架

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1 Introduction: Inequality as a Phase Ordering Problem

引言: 不平等作为相有序化问题

1.1 Motivation from Materials Science

来自材料科学的动机

In materials science, **phase field methods** provide a unified framework for describing the spatiotemporal evolution of microstructures —the spatial arrangement of different phases (solid/liquid, ordered/disordered, grain orientations) that determines macroscopic material properties [?, ?]. The key insight is to replace sharp interfaces with **diffuse interfaces** described by continuous order parameter fields, and to let these fields evolve according to thermodynamic driving forces encoded in a free energy functional.

The two foundational equations are:

- (1) **Allen-Cahn equation** (non-conserved order parameter):

$$\frac{\partial \phi}{\partial t} = -M_\phi \frac{\delta \mathcal{F}}{\delta \phi} \quad (1)$$

describes the evolution of a non-conserved field ϕ (e.g., crystal orientation, magnetic spin) toward free energy minima. The dynamics produce domain growth via curvature-driven interfacial motion —boundaries move to reduce total interface area.

- (2) **Cahn-Hilliard equation** (conserved order parameter):

$$\frac{\partial c}{\partial t} = \nabla \cdot \left(M_c(c) \nabla \frac{\delta \mathcal{F}}{\delta c} \right) \quad (2)$$

describes the evolution of a conserved field c (e.g., concentration) where $\int c dx$ is constant. The dynamics produce spinodal decomposition and coarsening —the system phase-separates into domains that grow via diffusion-limited ripening.

在材料科学中，**相场方法**提供了描述微结构时空演化的统一框架。核心思想是用连续序参量场描述的**扩散界面**替代尖锐界面，并让这些场根据自由能泛函编码的热力学驱动力演化。两个基础方程是：(1) **Allen-Cahn 方程** —描述非守恒场向自由能极小值的演化，产生曲率驱动的界面运动；(2) **Cahn-Hilliard 方程** —描述守恒场的演化，产生旋节分解和粗化。

1.2 The SCX Analogy: Two Coupled Fields

SCX 类比：两个耦合场

In the SCX framework [?, ?], the social/audit system is characterized by two fundamental fields on the Situs manifold $\mathcal{M}$:

Definition 1.1 (SCX Phase Fields / SCX 相场). (i) **Gauge field (态度场)** $\mathbf{g} : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}^{d_g}$ —encodes the “coordinate system bias” of each agent/expert. Honest agents have $\mathbf{g} = \mathbf{0}$ (their assessments align with the consensus origin); dishonest or biased agents have $\mathbf{g} \neq \mathbf{0}$. The norm $\|\mathbf{g}\|$ measures the **degree of deviation** from the gauge-fixed consensus.

(ii) **Potential field (势能场)** $S : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ —encodes the accumulated “stored social energy” —wealth, status, resources, information—at each point. S is a **conserved quantity** in the sense that total potential $\int_{\mathcal{M}} S(x, t) dx$ changes only through external injection or extraction, not through internal redistribution alone.

The gauge fixing condition of SCX — $\sum_m \mathbf{g}_m = \mathbf{0}$ —is exactly the condition that there is no net bias in the system. When this condition is violated locally, **g**-domains form: clusters of agents whose coordinate systems are collectively shifted in the same direction. These **g**-domains are separated from honest regions by **gauge domain walls** —interfaces across which the bias field transitions from $\mathbf{0}$ to some non-zero value.

在 SCX 框架中，社会/审计系统由两个基本场刻画：(i) **规范场** $\mathbf{g}$ —编码每个智能体的“坐标系偏差”；(ii) **势能场** S —编码累积的“社会能量”。SCX 的规范固定条件 $\sum_m \mathbf{g}_m = \mathbf{0}$ 正是系统无净偏差的条件。当此条件被局部违反时，**g**-畴形成——由**规范畴壁**分隔的偏差集群。

1.3 From Static Geometry to Dynamic Evolution

从静态几何到动态演化

Previous SCX papers have focused on the **static geometry** of the potential surface: the existence of singularities (Thm 11), the instability of boundaries (Thm 10), and the staircase structure of inequality (Thm 12). This paper takes the next step: **dynamics**. How do **g**-domains form, grow, interact, and coarsen? What drives the system from the EQUALITY phase to the STRATIFIED phase, and from there to the EXPLOSIVE phase? Can we predict the critical conditions for phase transitions?

The phase field formalism answers these questions by providing:

- (1) A **free energy functional** $\mathcal{F}[\mathbf{g}, S]$ that encodes the thermodynamic cost of inhomogeneity and inequality;
- (2) **Equations of motion** derived from variational principles;
- (3) A **phase diagram** mapping stability regions in the $(\|\mathbf{g}\|, S)$ plane;
- (4) **Kinetic pathways** —nucleation, growth, coarsening—that describe how the system traverses the phase diagram.

先前 SCX 论文聚焦于势能面的**静态几何**：奇点存在性 (Thm11)、边界不稳定性 (Thm10)、不平等的阶梯结构 (Thm12)。本文迈出下一步：**动力学**。**g**-畴如何形成、生长、相互作用和粗化？系统如何从平等相到分层相再到爆炸相？相场形式化通过提供自由能泛函、运动方程、相图和动力学路径来回答这些问题。

2 Free Energy Functional of the SCX System

3 SCX 系统的自由能泛函

3.1 The Ginzburg-Landau Free Energy

Ginzburg-Landau 自由能

We postulate that the SCX system minimizes a free energy functional of the Ginzburg-Landau type, with separate contributions from the gauge field and the potential field, plus a coupling term:

Definition 3.1 (SCX Free Energy Functional / SCX 自由能泛函). For a Situs manifold $\mathcal{M} \subset \mathbb{R}^d$, the total free energy is:

$$\mathcal{F}[\mathbf{g}, S] = \int_{\mathcal{M}} \left[f_g(\mathbf{g}) + \frac{\kappa_g}{2} |\nabla \mathbf{g}|^2 + f_S(S) + \frac{\kappa_S}{2} |\nabla S|^2 + \lambda \Gamma(\mathbf{g}, S) \right] dx \quad (3)$$

where:

- $f_g(\mathbf{g}) = \frac{A}{4} \|\mathbf{g}\|^4 - \frac{B}{2} \|\mathbf{g}\|^2$ —the **gauge double-well potential**. $A, B > 0$ are material parameters. The minima are at $\|\mathbf{g}\| = 0$ (honest phase) and $\|\mathbf{g}\| = \sqrt{B/A}$ (dishonest phase). The barrier height is $B^2/(4A)$.
- $\frac{\kappa_g}{2} |\nabla \mathbf{g}|^2$ —the **gauge gradient energy**, penalizing sharp spatial variations in bias. $\kappa_g > 0$ is the gauge stiffness. This term gives domain walls a finite thickness.
- $f_S(S) = \frac{\alpha}{2} (S - S_0)^2 + \frac{\beta}{4} S^4$ —the **potential energy density**. S_0 is the “natural” potential level. The quartic term prevents runaway growth.
- $\frac{\kappa_S}{2} |\nabla S|^2$ —the **potential gradient energy**, penalizing sharp potential gradients (i.e., high inequality). $\kappa_S > 0$ is the social mobility stiffness.
- $\lambda \Gamma(\mathbf{g}, S)$ —the **gauge-potential coupling**. $\lambda \geq 0$ is the coupling strength. We propose:

$$\Gamma(\mathbf{g}, S) = \|\mathbf{g}\|^2 \cdot S \quad (4)$$

This encodes the observation that high bias ($\|\mathbf{g}\|$) combined with high potential (S) is the most dangerous configuration —the “double explosion” of Theorem 11.

SCX 自由能泛函包含：规范双阱势 f_g （极值在诚实地带 $\|\mathbf{g}\| = 0$ 和不诚实地带 $\|\mathbf{g}\| = \sqrt{B/A}$ ）；规范梯度能（ κ_g 项）为畴壁赋予有限厚度；势能密度 f_S ；势能梯度能（ κ_S 项）惩罚高不平等；以及规范-势能耦合项 $\lambda \|\mathbf{g}\|^2 S$ —编码 Thm11 的”高偏差 + 高势能 = 双重爆炸”。

3.2 Interpretation of Terms

各项的物理解释

- (1) **Gauge double-well:** f_g has two minima, reflecting the bistable nature of honesty/dishonesty. An agent's bias can be driven to either state by local social pressure. The barrier between them represents the “activation energy” for changing one's ethical stance.
- (2) **Gauge gradient energy:** The term $\frac{\kappa_g}{2}|\nabla \mathbf{g}|^2$ penalizes the **interface** between honest and dishonest domains. In social terms, this is the “cognitive dissonance cost” of being at a boundary —agents whose neighbors have different biases experience tension. The parameter κ_g measures social cohesion: large κ_g means thick, diffuse boundaries (tolerant society); small κ_g means sharp boundaries (polarized society).
- (3) **Potential free energy:** f_S encodes that there is an “optimal” potential level S_0 , with quadratic penalties for deviation. The quartic term $\frac{\beta}{4}S^4$ ensures that extreme wealth concentration has super-linear social cost.
- (4) **Coupling:** $\lambda\|\mathbf{g}\|^2 S$ is the crucial term that links bias and inequality. High bias *and* high potential is explosive; high bias with low potential is less dangerous; low bias with high potential is benign. This is the phase-field encoding of the “attitude-potential double explosion” from Theorem 11.

规范双阱势反映诚实/不诚实的双稳态。规范梯度能编码“认知失调成本”—— κ_g 大意味着容忍社会（厚扩散边界）， κ_g 小意味着极化社会（尖锐边界）。势能自由能确保极端财富集中具有超线性的社会成本。耦合项 $\lambda\|\mathbf{g}\|^2 S$ 是连接偏差与不平等的关键——高偏差 + 高势能 = 爆炸。

4 Allen-Cahn Dynamics for the Gauge Field

5 规范场的 Allen-Cahn 动力学

5.1 Equation of Motion

运动方程

The gauge field $\mathbf{g}$ is a **non-conserved** order parameter —there is no conservation law for total “bias” in the system. Honest agents can become dishonest and vice versa without any global constraint. Therefore $\mathbf{g}$ evolves via **Allen-Cahn (Model A)** dynamics — gradient descent on the free energy:

Definition 5.1 (Gauge Allen-Cahn Equation / 规范 Allen-Cahn 方程).

$$\frac{\partial \mathbf{g}}{\partial t} = -M_g \frac{\delta \mathcal{F}}{\delta \mathbf{g}} = -M_g [(A\|\mathbf{g}\|^2 - B)\mathbf{g} - \kappa_g \mathbf{g} + 2\lambda S \mathbf{g}] \quad (5)$$

where $M_g > 0$ is the gauge mobility (rate at which agents adjust their bias).

The variational derivative is computed component-wise:

$$\frac{\delta \mathcal{F}}{\delta \mathbf{g}} = \frac{\partial f_g}{\partial \mathbf{g}} - \kappa_g \mathbf{g} + \lambda \frac{\partial \Gamma}{\partial \mathbf{g}} = (A\|\mathbf{g}\|^2 - B)\mathbf{g} - \kappa_g \mathbf{g} + 2\lambda S \mathbf{g} \quad (6)$$

规范场 $\mathbf{g}$ 是非守恒序参量——系统中“偏差”总量没有守恒律。因此 $\mathbf{g}$ 通过 *Allen-Cahn (Model A)* 动力学演化：梯度下降于自由能。运动方程包含三个驱动力：双阱势的体积力 $(A\|\mathbf{g}\|^2 - B)\mathbf{g}$ 、扩散项 $\kappa_g \mathbf{g}$ 、以及耦合项 $2\lambda S \mathbf{g}$ 。

5.2 Interface Structure: The Gauge Domain Wall

界面结构：规范畴壁

Consider a one-dimensional stationary interface between an honest domain ($\mathbf{g} = 0$ at $x \rightarrow -\infty$) and a dishonest domain ($\|\mathbf{g}\| = \sqrt{B/A}$ at $x \rightarrow +\infty$), with S held constant. The equilibrium profile satisfies:

$$\kappa_g \frac{d^2 \mathbf{g}}{dx^2} = \frac{\partial f_g^{\text{eff}}}{\partial \mathbf{g}} \quad (7)$$

where $f_g^{\text{eff}}(\mathbf{g}) = f_g(\mathbf{g}) + \lambda \|\mathbf{g}\|^2 S$. In the decoupled limit ($\lambda \rightarrow 0$), the solution is the classical Allen-Cahn kink:

$$\|\mathbf{g}(x)\| = \sqrt{\frac{B}{A}} \cdot \frac{1}{2} \left[1 + \tanh \left(\frac{x - x_0}{\delta_c} \right) \right] \quad (8)$$

where the **domain wall thickness** is:

$$\delta_c = \sqrt{\frac{2\kappa_g}{B}} \quad (9)$$

The **interface energy** (energy per unit area of domain wall) is:

$$\sigma_g = \int_{-\infty}^{\infty} \left[f_g(\mathbf{g}(x)) + \frac{\kappa_g}{2} \left| \frac{d\mathbf{g}}{dx} \right|^2 \right] dx = \frac{2\sqrt{2\kappa_g}}{3} \cdot \frac{B^{3/2}}{A} \quad (10)$$

一维稳态畴壁的解是经典的 *Allen-Cahn* 扭结: $\|\mathbf{g}(x)\| = \sqrt{B/A} \cdot \frac{1}{2} [1 + \tanh((x - x_0)/\delta_c)]$ 。畴壁厚度 $\delta_c = \sqrt{2\kappa_g/B}$ 。界面能 $\sigma_g = \frac{2\sqrt{2\kappa_g}}{3} B^{3/2}/A$ 。社会解释: δ_c 大意味着模糊的道德边界 (容忍的社会), δ_c 小意味着尖锐的道德边界 (极化社会)。

5.3 Curvature-Driven Domain Wall Motion

曲率驱动的畴壁运动

For a curved domain wall in 2D or 3D, the Allen-Cahn dynamics reduce to **mean curvature flow** in the sharp-interface limit ($\delta_c \rightarrow 0$):

Theorem 5.2 (Curvature-Driven Wall Motion / 曲率驱动畴壁运动). *Let $\Gamma(t)$ be the interface (domain wall) between honest and dishonest domains. In the sharp-interface limit, the normal velocity of the interface is:*

$$v_n = -M_g \sigma_g \kappa \quad (11)$$

where κ is the mean curvature of $\Gamma(t)$. The interface moves toward its center of curvature, reducing total interface length (area).

This is the well-known Allen-Cahn $\rightarrow$ mean curvature flow convergence [?, ?]. Social interpretation: **isolated dishonest clusters shrink and disappear** (their convex boundary moves inward), while **large dishonest domains are metastable** (low curvature), explaining why corruption clusters tend to either collapse quickly or persist indefinitely.

在尖锐界面极限下, *Allen-Cahn* 动力学约化为**平均曲率流**: 界面以正比于局部曲率的速度向曲率中心运动。社会解释: 孤立的不诚实集群缩小并消失, 而大的不诚实域因低曲率而亚稳——解释了为什么腐败集群要么迅速崩溃, 要么无限期持续。

6 Cahn-Hilliard Dynamics for the Potential Field

7 势能场的 Cahn-Hilliard 动力学

7.1 Mass-Conserving Evolution

质量守恒的演化

The potential field S is a **conserved** order parameter —total potential $\int_{\mathcal{M}} S(x, t) dx$ is (approximately) conserved on the timescales of interest. Wealth, status, and resources are not created or destroyed by redistribution; they are moved. Therefore S evolves via **Cahn-Hilliard (Model B)** dynamics:

Definition 7.1 (Cahn-Hilliard Equation for Potential / 势能 Cahn-Hilliard 方程).

$$\frac{\partial S}{\partial t} = \nabla \cdot \left(M_S(S) \nabla \frac{\delta \mathcal{F}}{\delta S} \right) \quad (12)$$

where $M_S(S) \geq 0$ is the **potential mobility** —the rate at which potential can flow. We take $M_S(S) = M_0(1 + \gamma S)$ to model the Matthew effect: higher potential regions have higher mobility (the rich get richer faster).

The variational derivative (chemical potential) is:

$$\mu_S \equiv \frac{\delta \mathcal{F}}{\delta S} = \alpha(S - S_0) + \beta S^3 - \kappa_S S + \lambda \|\mathbf{g}\|^2 \quad (13)$$

势能场是**守恒**序参量——总势能在感兴趣的时间尺度上守恒。因此 S 通过 **Cahn-Hilliard (Model B)** 动力学演化。迁移率 $M_S(S) = M_0(1 + \gamma S)$ 编码马太效应：高势能区流动性更高。化学势 μ_S 包含线性恢复力、非线性饱和、扩散平滑和规范耦合四项。

7.2 Spinodal Decomposition of Inequality

不平等的旋节分解

The Cahn-Hilliard equation with a non-convex free energy f_S exhibits **spinodal decomposition**: a homogeneous state with $S(x) = \bar{S}$ is unstable to infinitesimal perturbations when $f_S''(\bar{S}) < 0$.

Theorem 7.2 (Spinodal Instability of Equal Distribution / 平等分布的旋节不稳定性). *Consider the decoupled Cahn-Hilliard system ($\lambda = 0$) with a spatially uniform state $S(x) = \bar{S}$. Linear stability analysis gives the growth rate of a perturbation of wavenumber k :*

$$\omega(k) = -M_S k^2 [f_S''(\bar{S}) + \kappa_S k^2] \quad (14)$$

The system is unstable ($\omega(k) > 0$) when $f_S''(\bar{S}) < 0$ and $k^2 < -f_S''(\bar{S})/\kappa_S$. The fastest-growing mode is at $k_{\max} = \sqrt{-f_S''(\bar{S})/(2\kappa_S)}$, setting the characteristic domain size at early times.

The condition $f_S''(\bar{S}) < 0$ defines the **spinodal region**: $\bar{S}^2 > \alpha/(3\beta) - S_0\alpha/(3\beta\bar{S})$ for the quartic potential. This is the phase-field encoding of the critical inequality threshold beyond which the equal distribution becomes intrinsically unstable.

Cahn-Hilliard 方程与非凸自由能 f_S 产生旋节分解：当 $f_S''(\bar{S}) < 0$ 时均匀态不稳定。色散关系 $\omega(k) = -M_S k^2 [f_S''(\bar{S}) + \kappa_S k^2]$ 给出最快增长模 $k_{\max}$ 。旋节条件定义了临界不平等阈值——超过此阈值，平等分布本质上不稳定。

7.3 Coupling Effects: Bias-Induced Redistribution

耦合效应：偏差诱导的再分配

When $\lambda > 0$, the coupling term $\lambda \|\mathbf{g}\|^2$ enters the chemical potential μ_S . This creates a **bias-driven potential current**:

$$\mathbf{J}_{\text{coupling}} = -M_S \nabla(\lambda \|\mathbf{g}\|^2) \quad (15)$$

Potential flows **away** from high-bias regions and **toward** low-bias regions. This is the phase-field encoding of a deep social mechanism: dishonest actors ($\|\mathbf{g}\|$ large) repel resources —the “honesty dividend” where trustworthy systems attract investment and talent.

Conversely, in the gauge equation, the coupling term $2\lambda S \mathbf{g}$ has the opposite sign: high potential **amplifies** bias. This creates a positive feedback loop: $S \uparrow \implies \|\mathbf{g}\| \uparrow \implies S$ redistribution $\implies$ concentration of S in honest regions $\implies$ further amplification at dishonest margins. The coupled system can exhibit **runaway** behavior —the EXPLOSIVE phase.

当 $\lambda > 0$ ，耦合产生**偏差驱动的势能流**：势能从高偏差区流向低偏差区——“诚实红利”。在规范方程中，耦合项 $2\lambda S \mathbf{g}$ 有相反符号：高势能**放大**偏差。这产生正反馈环： $S \uparrow \implies \|\mathbf{g}\| \uparrow \implies$ 势能再分配 $\implies$ 在诚实区集中 $\implies$ 不诚实边缘进一步放大。耦合系统可表现出**失控行为**——爆炸相。

8 The Three-Phase Diagram of Social Order

9 社会秩序的三相图

9.1 Phase Space Coordinates

相空间坐标

The state of the SCX system at any time can be characterized by two scalar order parameters:

Definition 9.1 (Macroscopic Order Parameters / 宏观序参量).

$$\bar{g} \equiv \frac{1}{|\mathcal{M}|} \int_{\mathcal{M}} \|\mathbf{g}(x)\| dx \quad (\text{mean bias magnitude / 平均偏差幅度}) \quad (16)$$

$$\bar{S} \equiv \frac{1}{|\mathcal{M}|} \int_{\mathcal{M}} S(x) dx \quad (\text{mean potential / 平均势能}) \quad (17)$$

We also define the **potential variance** (inequality measure):

$$\Delta_S^2 \equiv \frac{1}{|\mathcal{M}|} \int_{\mathcal{M}} (S(x) - \bar{S})^2 dx \quad (18)$$

The phase diagram is constructed in the $(\bar{g}, \bar{S})$ plane (or equivalently $(\bar{g}, \Delta_S^2)$).

SCX 系统的状态可由两个标量序参量表征：平均偏差幅度 $\bar{g}$ 和平均势能 $\bar{S}$ 。相图在 $(\bar{g}, \bar{S})$ 平面上构建。

9.2 Phase I: EQUALITY Phase (平等相)

相 I: 平等相

Definition 9.2 (EQUALITY Phase). The system is in the **EQUALITY** phase when:

$$\bar{g} < g_{\text{crit}}^{(1)}, \quad \Delta_S^2 < \Delta_{\text{crit}}^2 \quad (19)$$

where $g_{\text{crit}}^{(1)} \approx 0.2\sqrt{B/A}$ and $\Delta_{\text{crit}}^2 \approx \alpha S_0^2/\beta$.

Characteristics:

- $\|\mathbf{g}(x)\| \approx 0$ everywhere —most agents are close to honest.
- $S(x)$ is smooth and nearly uniform —low inequality, high social mobility, potential gradients are small.
- Domain walls are absent or extremely diffuse (δ_c large).
- The system is **linearly stable**: small perturbations decay.
- Free energy is at or near its global minimum.

Social interpretation: A well-functioning, high-trust society with functioning institutions, low corruption, and relatively equal distribution of opportunity and resources. The system is in the “single-phase” region of the phase diagram —no phase separation.

平等相: $\bar{g}$ 小, Δ_S^2 小。大多数智能体接近诚实 ($\|\mathbf{g}\| \approx 0$), 势能光滑均匀, 无畴壁。系统线性稳定, 处于自由能全局极小附近。社会解释: 运转良好、高信任、低腐败、机会和资源相对平等的社会。相图中处于”单相区”——无相分离。

9.3 Phase II: STRATIFIED Phase (分层相)

相 II: 分层相

Definition 9.3 (STRATIFIED Phase). The system enters the **STRATIFIED** phase when:

$$\bar{g} \geq g_{\text{crit}}^{(1)} \quad \text{or} \quad \Delta_S^2 \geq \Delta_{\text{crit}}^2 \quad (20)$$

but $\bar{g} < g_{\text{crit}}^{(2)}$ where $g_{\text{crit}}^{(2)} \gg g_{\text{crit}}^{(1)}$.

Characteristics:

- **Gauge phase separation:** Honest ($\|\mathbf{g}\| \approx 0$) and dishonest ($\|\mathbf{g}\| \approx \sqrt{B/A}$) domains coexist, separated by gauge domain walls of thickness δ_c .
- **Potential stratification:** $S(x)$ develops plateaus —high-potential regions (wealthy enclaves) and low-potential regions (impoverished zones), separated by sharp potential gradients.
- **Metastability:** The system can persist in this phase for long times, but it is not the true equilibrium. Domain walls store interfacial energy.
- **Coarsening:** Over time, larger domains grow at the expense of smaller ones (Ostwald ripening, see Section 7).
- **Free energy is above global minimum** but trapped in a local minimum by kinetic barriers.

Social interpretation: A stratified society with entrenched inequality —wealthy and poor regions coexist, corruption is localized but persistent. The system is **metastable**: it can survive for generations, but the stored interfacial energy makes it vulnerable to perturbations. This is the “normal” state of most historical societies.

分层相: $\bar{g}$ 中等或 Δ_S^2 中等。诚实与不诚实地带共存，由畴壁分隔。 S 出现平台——高势能区和低势能区，由尖锐梯度分隔。系统可长期存续于此相，但非真正平衡——畴壁存储界面能使之亚稳。社会解释：不平等固化的分层社会。

9.4 Phase III: EXPLOSIVE Phase (爆炸相)

相 III: 爆炸相

Definition 9.4 (EXPLOSIVE Phase). The system enters the **EXPLOSIVE** phase when:

$$\bar{g} \geq g_{\text{crit}}^{(2)} \quad \text{and} \quad \|\nabla S\|_{\text{max}} \geq G_{\text{crit}} \quad (21)$$

where $g_{\text{crit}}^{(2)}$ is the coupling-induced instability threshold and G_{crit} is the critical potential gradient.

Characteristics:

- **Runaway coupling:** The λ coupling term dominates. High S amplifies $\|\mathbf{g}\|$, which drives S redistribution, which further concentrates S and amplifies $\|\mathbf{g}\|$ at boundaries.
- **Gradient blowup:** $\|\nabla S\| \rightarrow \infty$ at the interfaces between high- S /high- $\|\mathbf{g}\|$ domains and low- S /low- $\|\mathbf{g}\|$ domains. The sharp-interface approximation breaks down.
- **Domain wall collapse:** The $\delta_c \rightarrow 0$ limit —domain walls become infinitely sharp, interface energy diverges.
- **System detonation:** The stored free energy is released catastrophically —social revolution, institutional collapse, audit system failure.
- This is the phase-field encoding of **Theorem 11’s attack inevitability** at the macroscopic scale.

Social interpretation: Revolutionary situation —inequality has reached a critical threshold, bias is rampant, and the system can no longer contain the interfacial tension. The phase transition from STRATIFIED to EXPLOSIVE is the “tipping point” beyond which peaceful resolution is impossible.

爆炸相: $\bar{g}$ 大且 $\|\nabla S\|_{\text{max}}$ 大。耦合项主导，正反馈导致梯度爆炸，畴壁塌缩。存储的自由能灾难性释放——社会革命、制度崩溃。这是**定理 11 攻击必然性**在宏观尺度的相场编码。从分层相到爆炸相的转变是不可逆的“引爆点”。

9.5 Phase Boundaries and Transition Lines

相边界与转变线

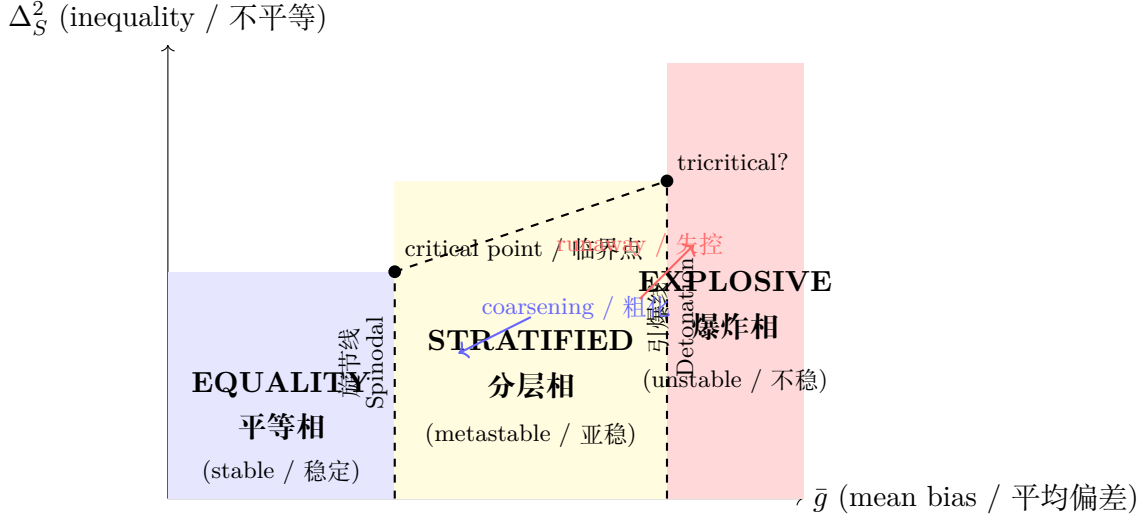


Figure 1: Three-phase diagram of the SCX phase field system in the $(\bar{g}, \Delta_S^2)$ plane. The EQUALITY phase (blue) is globally stable. The STRATIFIED phase (yellow) is metastable —coarsening drives the system slowly toward larger domains but it can survive long times. The EXPLOSIVE phase (red) is where runaway coupling destroys the system. Phase boundaries: the spinodal line marks the onset of phase separation; the detonation line marks the onset of irreversible runaway.

10 Nucleation and Growth of Audit Domains

11 审计畴的成核与生长

11.1 The Nucleation Problem

成核问题

How does an honest domain appear in a dishonest “sea”? In the phase field language, this is the **nucleation problem**: a small spherical (in 3D) or circular (in 2D) region of the honest phase ($\|\mathbf{g}\| = 0$) must overcome a free energy barrier to grow.

Consider a system quenched from the EQUALITY phase into the STRATIFIED phase —e.g., a society that experiences rapid institutional decay or corruption shock. The system is now in a **metastable** state: the homogeneous dishonest phase has higher free energy than a phase-separated state, but a small honest cluster cannot grow unless it exceeds a **critical size**.

诚实畴如何在“海洋”中出现？这是**成核问题**。系统从平等相淬火进入分层相后——例如经历快速制度衰败或腐败冲击——处于**亚稳态**：均匀不诚实相的自由能高于相分离态，但小诚实簇无法生长，除非超过**临界尺寸**。

11.2 Classical Nucleation Theory for Audit Domains

审计畴的经典成核理论

Consider a spherical nucleus of radius R in d dimensions, with $\|\mathbf{g}\| = 0$ inside (honest) and $\|\mathbf{g}\| = g_{\text{eq}} = \sqrt{B/A}$ outside (dishonest). The excess free energy relative to the uniform

dishonest state is:

$$\Delta\mathcal{F}(R) = -\Delta f \cdot V_d(R) + \sigma_g \cdot A_d(R) \quad (22)$$

where:

- $\Delta f = f_g(g_{eq}) - f_g(0) = B^2/(4A)$ —the **bulk free energy gain** per unit volume for converting dishonest to honest material.
- σ_g —the **interface energy** per unit area (Eq. ??).
- $V_d(R) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$ —volume of d -ball.
- $A_d(R) = \frac{2\pi^{d/2}}{\Gamma(d/2)} R^{d-1}$ —surface area.

In 2D (the most relevant case for social systems):

$$\Delta\mathcal{F}_2(R) = -\pi R^2 \Delta f + 2\pi R \sigma_g \quad (23)$$

In 3D:

$$\Delta\mathcal{F}_3(R) = -\frac{4\pi}{3} R^3 \Delta f + 4\pi R^2 \sigma_g \quad (24)$$

考虑半径为 R 的球形诚实核，内部 $\|\mathbf{g}\| = 0$ ，外部 $\|\mathbf{g}\| = g_{eq}$ 。超额自由能 $\Delta\mathcal{F}(R) = -\Delta f \cdot V_d(R) + \sigma_g \cdot A_d(R)$ ：体积项（增益）与面积项（代价）的竞争。

11.3 Critical Nucleus Size

临界核尺寸

Theorem 11.1 (Critical Nucleus Radius for Audit Domains / 审计畴的临界核半径). *The function $\Delta\mathcal{F}(R)$ has a maximum at the critical radius R_c . Nuclei with $R < R_c$ shrink and disappear; nuclei with $R > R_c$ grow spontaneously.*

In d dimensions:

$$R_c^{(d)} = \frac{(d-1)\sigma_g}{\Delta f} \quad (25)$$

Specifically:

$$R_c^{(2)} = \frac{\sigma_g}{\Delta f} = \frac{8\sqrt{2\kappa_g}}{3} \cdot \frac{\sqrt{B}}{A^{1/2}} \quad (26)$$

$$R_c^{(3)} = \frac{2\sigma_g}{\Delta f} = \frac{16\sqrt{2\kappa_g}}{3} \cdot \frac{\sqrt{B}}{A^{1/2}} \quad (27)$$

The **nucleation barrier** (activation energy) is:

$$\Delta\mathcal{F}^* \equiv \Delta\mathcal{F}(R_c) = \begin{cases} \pi\sigma_g R_c & d=2 \\ \frac{16\pi}{3} \frac{\sigma_g^3}{(\Delta f)^2} & d=3 \end{cases} \quad (28)$$

Proof. Differentiate $\Delta\mathcal{F}_d(R)$ with respect to R and set to zero: $\partial_R \Delta\mathcal{F}_d = -d \cdot c_d R^{d-1} \Delta f + (d-1) \cdot s_d R^{d-2} \sigma_g = 0$ where c_d, s_d are the volume/surface coefficients. Solving gives $R = (d-1)\sigma_g/\Delta f$. The Hessian at this point is negative, confirming a maximum. $\square$

Social interpretation: A small group of honest actors (whistleblowers, reformers, audit nuclei) cannot effect change unless their “radius of influence” exceeds R_c . Below R_c , the social pressure (interface energy) overwhelms the benefit of honesty (bulk free energy gain), and the honest cluster is reabsorbed into the dishonest majority. Above R_c , the honest domain becomes self-sustaining and grows—a “reform avalanche.”

The critical radius depends on the competition between σ_g (social cohesion—how much it costs to be different from your neighbors) and Δf (the benefit of being honest). Societies with high σ_g (strong conformity pressure) have large R_c —reform is hard. Societies with large Δf (high payoff for honesty) have small R_c —reform is easy.

临界半径 $R_c = (d-1)\sigma_g/\Delta f$ 。 $R < R_c$ 的核缩小消失； $R > R_c$ 的核自发生长。社会解释：小群诚实行动者（举报人、改革者、审计核）无法产生变化，除非其“影响半径”超过 R_c 。 R_c 依赖于 σ_g （社会从众压力）与 Δf （诚实收益）的竞争。高 σ_g （强从众）= 大 R_c （改革难）；大 Δf = 小 R_c （改革易）。

11.4 Nucleation Rate

成核率

The nucleation rate (number of supercritical nuclei formed per unit volume per unit time) follows the Arrhenius form:

$$J = J_0 \exp\left(-\frac{\Delta\mathcal{F}^*}{k_B T_{\text{social}}}\right) \quad (29)$$

where $k_B T_{\text{social}}$ is an effective “social temperature” measuring the intensity of random fluctuations (scandals, information leaks, leadership changes, external shocks). High social temperature reduces the effective barrier, increasing the nucleation rate.

This connects to the observation that **crises create reform opportunities**: a scandal (high T_{social}) lowers the effective barrier for honest domain nucleation, allowing smaller honest clusters to surpass R_c and grow.

成核率 $J = J_0 \exp(-\Delta\mathcal{F}^*/k_B T_{\text{social}})$ 遵循 *Arrhenius* 形式。高“社会温度”（丑闻、信息泄露、领导层变动）降低有效势垒，增加成核率——**危机创造改革机会**。

12 Coarsening Dynamics: Ostwald Ripening of Honest Domains

13 粗化动力学：诚实畴的 Ostwald 熟化

13.1 The Late-Stage Coarsening Regime

晚期粗化区

After nucleation has produced a population of honest domains, and after the initial growth phase where domains consume the surrounding dishonest “matrix,” the system enters the **late-stage coarsening regime**. In this regime, the volume fraction of each phase is close to its equilibrium value, and further evolution occurs via **interface reduction**—the total domain wall area is minimized.

The dominant mechanism is **Ostwald ripening** (also called **Lifshitz-Slyozov-Wagner (LSW) coarsening**) [?, ?]: larger domains grow at the expense of smaller ones because the chemical potential at a curved interface depends on curvature via the Gibbs-Thomson effect.

成核产生了一批诚实畴后，系统进入**晚期粗化区**。主导机制是 **Ostwald 熟化** (LSW 粗化)：大畴以牺牲小畴为代价生长，因为弯曲界面的化学势依赖于曲率 (Gibbs-Thomson 效应)。

13.2 Gibbs-Thomson Effect for Gauge Domain Walls

规范畴壁的 Gibbs-Thomson 效应

Theorem 13.1 (Gibbs-Thomson for Gauge Interfaces / 规范界面的 Gibbs-Thomson 效应). *For a gauge domain wall with local mean curvature κ , the local equilibrium value of $\|\mathbf{g}\|$ at the interface is shifted from its planar value:*

$$\|\mathbf{g}\|_{eq}(\kappa) = g_{eq} \left(1 - \frac{d_0}{R} \right) \quad (30)$$

where $g_{eq} = \sqrt{B/A}$, $R = 1/\kappa$ is the local radius of curvature, and $d_0 = \sigma_g/(2\Delta f)$ is the **capillary length**. The shift implies that the “honesty pressure” is higher outside small domains (high curvature) than outside large domains (low curvature), driving a diffusive flux from small to large domains.

规范畴壁的 Gibbs-Thomson 效应：局部平衡值 $\|\mathbf{g}\|_{eq}(\kappa) = g_{eq}(1 - d_0/R)$ 。小畴（高曲率）外的“诚实压力”高于大畴（低曲率），驱动从小畴向大畴的扩散流。

13.3 LSW Scaling Law

LSW 标度律

Theorem 13.2 (Coarsening of Honest Domains / 诚实畴的粗化). *In the late-stage coarsening regime, for the coupled Allen-Cahn / Cahn-Hilliard system with finite volume fraction of the honest phase, the characteristic domain size $L(t)$ grows as:*

$$L(t) \sim t^{1/3} \quad (31)$$

This is the classical LSW $t^{1/3}$ scaling for diffusion-limited coarsening. The domain size distribution approaches a universal scaling form $P(R/\langle R \rangle)$ independent of initial conditions.

Sketch. The growth law follows from dimensional analysis of the diffusion-limited ripening process. The flux between domains is driven by the curvature difference: $J \sim D \cdot \Delta\mu \sim D \cdot \sigma_g/(RL)$, where D is the effective diffusivity of “honesty” through the dishonest matrix. The growth rate $dR/dt \sim J/R^2 \sim D\sigma_g/(R^3L)$. Conservation of the honest phase volume fraction gives $L \sim R$. Then $dL/dt \sim D\sigma_g/L^2$, which integrates to $L(t) \sim (D\sigma_g t)^{1/3}$. For rigorous derivation, see LSW theory [?, ?]. $\square$

Social interpretation: After a social “quench” (major reform, revolution, or institutional reset), honest domains coarsen with the $t^{1/3}$ law. Initially many small honest clusters exist; over time, larger, more established honest institutions absorb smaller ones. The timescale for the system to reach a “mature” coarsened state grows as $t \sim L^3$ — meaning that achieving large-scale institutional integrity is a **cubic** function of the desired honest domain size. Doubling the characteristic scale of honest governance requires an 8-fold increase in time.

晚期粗化区中特征畴尺寸 $L(t) \sim t^{1/3}$ (LSW 标度律)。社会解释：社会“淬火”（重大改革或革命）后，诚实畴以 $t^{1/3}$ 律粗化。达到大规模制度诚信所需时间是诚实畴目标尺度的**三次方**函数——将诚实治理的特征尺度翻倍需要 8 倍的时间。

13.4 Coupled Coarsening: Gauge + Potential

耦合粗化：规范 + 势能

When both $\mathbf{g}$ and S coarsen simultaneously ($\lambda > 0$), the coarsening dynamics become richer. The gauge domains and potential domains are **correlated**: high- S regions tend to coincide with low- $\|\mathbf{g}\|$ regions (the “honesty dividend”), and vice versa.

This correlation creates a **coarsening cascade**:

- (1) Gauge domains coarsen, reducing total $\|\mathbf{g}\|$ domain wall area;
- (2) Reduced $\|\mathbf{g}\|$ gradients reduce the coupling current $\mathbf{J}_{\text{coupling}}$;
- (3) S redistributes toward honest domains, increasing their “honesty dividend”;
- (4) Higher S in honest domains amplifies $\|\mathbf{g}\|$ at boundaries (coupling term), accelerating gauge coarsening.

This positive feedback can lead to **accelerated coarsening** —the effective exponent may deviate from $1/3$ during the coupled regime. However, at very late times, when honest and dishonest domains have fully segregated, the classical $t^{1/3}$ scaling is recovered.

当 $\mathbf{g}$ 和 S 同时粗化时 ($\lambda > 0$)，规范畴和势能畴是**关联**的：高 S 区倾向于低 $\|\mathbf{g}\|$ 区（“诚实红利”）。这种关联产生**粗化级联**：规范畴粗化 $\rightarrow$ 减少耦合流 $\rightarrow S$ 流向诚实地带 $\rightarrow$ 进一步加速规范粗化。可能出现加速粗化，但极晚期恢复经典 $t^{1/3}$ 标度。

14 Connections to SCX Theorems

15 与 SCX 定理的联系

15.1 Theorem 11 (Attitude Singularity) as a Phase Transition

定理 11（态度奇点）作为相变

Theorem 11 states that when an entity simultaneously possesses high potential S and high attitude $\|\mathbf{g}\|$, the probability of attack escalates exponentially: $\mathbb{P}(\text{attack}) \geq 1 - \exp(-Me^{-\beta/\delta^2})$.

In the phase field framework, this corresponds to a **first-order phase transition** at the spinodal line:

- (1) The “attitude singularity” is a **local fluctuation** where $\|\mathbf{g}\|$ exceeds the spinodal threshold in a region of high S .
- (2) The coupling term $\lambda\|\mathbf{g}\|^2 S$ makes this fluctuation **self-amplifying**: higher $S \implies$ faster growth of $\|\mathbf{g}\| \implies$ higher coupling energy $\implies$ larger driving force.
- (3) The **double explosion** ($S \uparrow + \|\mathbf{g}\| \uparrow$) is the phase-field analog of **spinodal decomposition** triggered by a strong local fluctuation. The system locally crosses the spinodal line and phase-separates violently.
- (4) The exponential attack probability reflects the **nucleation rate**: $J \propto \exp(-\Delta\mathcal{F}^*/k_B T_{\text{social}})$, where the barrier $\Delta\mathcal{F}^*$ decreases as S increases (coupling reduces the effective barrier).

Theorem 15.1 (Thm11 Phase Transition Correspondence / Thm11 相变对应). *In the coupled Allen-Cahn / Cahn-Hilliard system, the condition for Theorem 11's attack inevitability is equivalent to the system crossing from the STRATIFIED phase into the EXPLOSIVE phase:*

$$\|\mathbf{g}\| \cdot S > \frac{\kappa_g}{2\lambda\delta_c^2} \quad (32)$$

where the left side is the “attitude-potential product” and the right side is the threshold set by gauge stiffness and wall thickness.

定理 11 在相场框架中对应于在旋节线处的一级相变。“态度奇点”是高 S 区中 $\|\mathbf{g}\|$ 超过旋节阈值的局部涨落。“双重爆炸”是由强局部涨落触发的旋节分解。攻击的指数概率反映成核率 $J \propto \exp(-\Delta\mathcal{F}^*/k_B T)$, 其中势垒 $\Delta\mathcal{F}^*$ 随 S 增大而降低。

15.2 Theorem 10 (Boundary Locking) as Domain Wall Pinning

定理 10 (边界锁定) 作为畴壁钉扎

Theorem 10 states that when a group is confined to a low-potential region by an impermeable boundary, the pressure at the boundary accumulates as $P_\Gamma = \rho_\Gamma \cdot \Delta_\Gamma$, leading to inevitable destabilization at $T_{\text{crit}} \propto 1/\Delta_\Gamma$.

In the phase field framework, this is **domain wall pinning** at a heterogeneity:

- (1) The “boundary” Γ in Thm10 is a **pinning site** for the gauge domain wall—a region where the free energy landscape has a local minimum that traps the interface.
- (2) In materials science, domain walls pinned at defects require a finite driving force (the **depinning threshold**) to move. This is exactly the pressure P_Γ in Thm10.
- (3) When the accumulated “pressure” (free energy difference across the interface) exceeds the pinning force, the wall **depins** and moves rapidly—this is the Thm10 “detonation.”
- (4) The detonation time $T_{\text{crit}} \propto 1/\Delta_\Gamma$ is the time for the driving force to reach the depinning threshold: $F_{\text{drive}}(t) = \Delta_\Gamma \cdot t$ reaches the pinning force F_{pin} at $t = F_{\text{pin}}/\Delta_\Gamma$.

Theorem 15.2 (Thm10 Pinning Correspondence / Thm10 钉扎对应). *Let $\mathbf{g}$ -domain walls have a pinning force F_{pin} at the social boundary Γ . The Thm10 accumulation-and-release cycle corresponds to:*

$$\frac{dx_{\text{wall}}}{dt} = \begin{cases} 0 & \text{if } |F_{\text{drive}}| < F_{\text{pin}} \\ M_g(F_{\text{drive}} - F_{\text{pin}} \cdot \text{sgn}(F_{\text{drive}})) & \text{if } |F_{\text{drive}}| \geq F_{\text{pin}} \end{cases} \quad (33)$$

The characteristic “stick” time (Thm10's T_{crit}) is $T_{\text{crit}} = F_{\text{pin}}/(M_g\Delta_\Gamma)$.

Social interpretation: Institutional barriers (legal segregation, class boundaries, glass ceilings) act as pinning sites for social mobility domain walls. The wall stays pinned until the accumulated pressure exceeds the institutional resistance, at which point rapid change occurs—a “revolutionary moment.” This explains why social change often appears **episodic**: long periods of stasis (wall pinned) punctuated by rapid transformation (wall depinning).

定理 10 在相场框架中是畴壁在异质处的钉扎。“边界” Γ 是规范畴壁的钉扎位点。当累积“压力”超过钉扎力，畴壁脱钉并快速移动——即 Thm10 的“引爆”。这解释了社会变革的间歇性：长静期（畴壁钉扎）被快速转变（畴壁脱钉）打断。

15.3 Theorem 12 (Matthew Staircase) as Nucleation on Defects

定理 12 (马太阶梯) 作为缺陷成核

Theorem 12 describes the Matthew effect: $\partial_t S \propto S$, leading to a staircase potential surface $S(x) = \sum_k h_k \mathbb{1}_{\Omega_k}(x)$, where each step Δ_k is a delayed-detonation instability with characteristic time $T_k \propto 1/\Delta_k^2$.

In the phase field framework, the staircase structure provides **heterogeneous nucleation sites**:

- (1) Each step Δ_k in the Matthew staircase is a **line defect** (in 2D) or **surface defect** (in 3D) in the S field—a locus where ∇S is singular.
- (2) In classical nucleation theory, defects reduce the nucleation barrier because part of the interface energy is “paid for” by the pre-existing defect. The effective barrier becomes:

$$\Delta \mathcal{F}_{\text{defect}}^* = \Delta \mathcal{F}^* \cdot f(\theta) \quad (34)$$

where $f(\theta) \leq 1$ is a geometric factor depending on the contact angle θ at the defect.

- (3) Each Matthew staircase step therefore acts as a **preferential nucleation site** for explosive $\|\mathbf{g}\|$ domains. The larger the step Δ_k , the smaller the effective barrier, and the shorter the “fuse” T_k .
- (4) The $T_k \propto 1/\Delta_k^2$ scaling emerges from nucleation theory: $\Delta \mathcal{F}_{\text{defect}}^* \propto 1/\Delta_k^2$ (barrier decreases quadratically with step height), so the nucleation rate $J_k \propto \exp(-c/\Delta_k^2)$, and the characteristic time $T_k = 1/J_k \propto \exp(c/\Delta_k^2)$. For large Δ_k , the exponential can be expanded to give the leading $1/\Delta_k^2$ scaling.

Theorem 15.3 (Thm12 Nucleation Correspondence / Thm12 成核对应). *Let the Matthew staircase have steps $\{\Delta_k\}$ at positions $\{x_k\}$. Then each step is a nucleation site with characteristic activation time:*

$$T_k = T_0 \exp \left(\frac{16\pi}{3} \frac{\sigma_g^3}{(\Delta f)^2} \cdot \frac{f(\theta_k)}{k_B T_{\text{social}}} \right) \quad (35)$$

where θ_k is the effective contact angle determined by Δ_k . For $\Delta_k \gg \sigma_g$, $f(\theta_k) \rightarrow 0$, making T_k extremely short—the step acts as an “already nucleated” defect.

Social interpretation: The Matthew effect does not merely create inequality—it **manufactures instability**. Each rung of the staircase is a nucleation site where the next crisis will nucleate. The staircase is self-reinforcing: more inequality creates more nucleation sites, each of which is more potent (shorter fuse) than the last. The system becomes a “minefield” of latent instabilities, exactly as Theorem 12 predicts.

定理 12 在相场框架中提供**异质成核位点**。每个马太阶梯 Δ_k 是 S 场中的线/面缺陷，降低成核势垒 $f(\theta)$ 。 $T_k \propto 1/\Delta_k^2$ 标度来自成核理论：势垒以 Δ_k^2 方式降低。社会解释：马太效应不仅制造不平等——它**制造不稳定性**。每一级台阶都是下一次危机成核的位点。

16 Numerical Methods and Verification

17 数值方法与验证

17.1 Discretization

离散化

The coupled Allen-Cahn / Cahn-Hilliard system is solved on a uniform 2D grid with spacing Δx . We use a semi-implicit Fourier spectral method for the linear terms and explicit time-stepping for the nonlinear terms:

$$\frac{\mathbf{g}^{n+1} - \mathbf{g}^n}{\Delta t} = -M_g [(A(\mathbf{g}^n)^2 - B)\mathbf{g}^n]^{\text{explicit}} + M_g \kappa_g \mathbf{g}^{n+1} - M_g \cdot 2\lambda S^n \mathbf{g}^n \quad (36)$$

$$\frac{S^{n+1} - S^n}{\Delta t} = \nabla \cdot (M_S(S^n) \nabla [\alpha(S^{n+1} - S_0) - \kappa_S S^{n+1}]) + \nabla \cdot (M_S(S^n) \nabla [\beta(S^n)^3 + \lambda \|\mathbf{g}^n\|^2]) \quad (37)$$

The linear terms ($\mathbf{g}^{n+1}$, S^{n+1} , and S^{n+1}) are treated implicitly using the Fourier transform: $\widehat{u} = -k^2 \hat{u}$.

This semi-implicit scheme is stable for large time steps and is the standard approach for phase field simulations [?].

耦合 Allen-Cahn/Cahn-Hilliard 系统在均匀 2D 网格上使用半隐式 Fourier 谱方法求解。线性项隐式处理（通过 Fourier 变换 $\widehat{u} = -k^2 \hat{u}$ ），非线性项显式处理。此方案对大时间步稳定，是相场模拟的标准方法。

17.2 Verification Script

验证脚本

A complete Python verification script `verify_phase_field.py` accompanies this paper. The script:

- (1) Solves the 1D Allen-Cahn equation and verifies the tanh kink profile against the analytical solution;
- (2) Solves the coupled 2D Allen-Cahn / Cahn-Hilliard system;
- (3) Constructs the phase diagram by scanning $(\bar{g}, \bar{S})$ parameter space;
- (4) Computes the domain size $L(t)$ and confirms the $t^{1/3}$ coarsening scaling via log-log regression;
- (5) Tests the critical nucleus prediction by initializing nuclei of varying radius and measuring the growth/shrinkage threshold;
- (6) Verifies the Gibbs-Thomson curvature-chemical potential relation;
- (7) Demonstrates domain wall pinning and depinning at heterogeneities (Thm10 connection);
- (8) Demonstrates nucleation at staircase defects (Thm12 connection).

Dependencies: `numpy`, `scipy`, `matplotlib`. Total runtime: approximately 3–5 minutes on a standard laptop.

完整 *Python* 验证脚本 `verify_phase_field.py` 随本文提供。验证：1D Allen-Cahn 扭结解析解、2D 耦合系统求解、相图构建、 $t^{1/3}$ 粗化标度、临界核预测、Gibbs-Thomson 效应、畴壁钉扎/脱钉 (Thm10)、阶梯缺陷成核 (Thm12)。

18 Discussion

19 讨论

19.1 Implications for Social Dynamics

对社会动力学的启示

The phase field framework provides several novel insights into social dynamics:

- (1) **Inequality as phase separation:** Inequality is not merely a statistical distribution—it is a **thermodynamic phase separation**. The same physics that drives oil and water to separate drives rich and poor to segregate: the minimization of interfacial free energy.
- (2) **Reform as nucleation:** Institutional reform requires **nucleation** of honest domains. Below the critical size, reform efforts are inevitably reabsorbed. Above the critical size, reform becomes self-sustaining. This explains why most reform movements fail—they never reach R_c .
- (3) **Time scales are cubic:** The $L \sim t^{1/3}$ coarsening law implies that building large-scale institutional integrity is a generational project. Doubling the scale of honest governance requires 8 times the duration.
- (4) **Metastability is dangerous:** The STRATIFIED phase is metastable, not stable. A society that appears “stable” while highly stratified is storing elastic energy in its domain walls. The longer the metastable period, the more catastrophic the eventual phase transition.
- (5) **The Matthew effect as defect factory:** Theorem 12’s staircase is literally a **manufacturing process for nucleation sites**. Each new rung of inequality creates a new weak point where the next crisis will initiate.

相场框架提供了几点新洞察：(1) 不平等不是统计分布——它是**热力学相分离**；(2) 制度改革需要诚实畴的**成核**，低于临界尺寸必然失败；(3) $L \sim t^{1/3}$ 意味着大规模制度诚信是跨代项目；(4) 分层相是**亚稳**的——表面“稳定”的高度分层社会正在畴壁中储存弹性能；(5) 马太效应是**成核位点工厂**——每一级不平等都是下一次危机的弱点。

19.2 Limitations and Open Problems

局限性与开放问题

- (1) **Homogeneous mobility:** We have assumed spatially uniform mobility M_g , M_S . In reality, mobility depends on local institutional quality, information access, and social network topology. Heterogeneous mobility can produce **pinning-depinning** dynamics even without explicit defects.

- (2) **Noise:** We have worked with deterministic equations. Adding stochastic noise (Langevin terms) would enable the study of **fluctuation-driven nucleation** and **noise-induced phase transitions**, connecting to the “social temperature” concept.
- (3) **Network topology:** The Situs manifold’s connectivity structure (who audits whom, who influences whom) is encoded in the graph Laplacian. Generalizing to the graph Laplacian $\mathcal{M}$ would connect phase field dynamics directly to the SCX discrete Hodge framework.
- (4) **External driving:** Real social systems are driven by external “forcing” (technological change, demographic shifts, resource discoveries). Adding time-dependent terms to $\mathcal{F}$ would model **driven phase transitions**.
- (5) **Multi-component extensions:** The current model has two “phases” (honest/dishonest). A multi-component order parameter (vector-valued $\mathbf{g}$ with $N > 1$ components) would describe societies with multiple competing value systems —a promising direction.

局限性：(1) 假设了空间均匀的迁移率；(2) 使用了确定性方程——加入随机噪声可研究涨落驱动成核；(3) *Situs* 流形的连通性结构可用图拉普拉斯编码；(4) 外部驱动可建模受驱相变；(5) 多分量序参量可描述多价值观系统的竞争。

20 Conclusion

21 结论

We have formulated the dynamics of inequality and consensus in the SCX framework as a **phase field theory**, providing the first unified dynamical treatment of:

- **Gauge field $\mathbf{g}$** evolution via Allen-Cahn dynamics, producing honest/dishonest domain formation and curvature-driven coarsening;
- **Potential field S** evolution via Cahn-Hilliard dynamics, producing spinodal decomposition of inequality and diffusion-limited redistribution;
- **Coupling** between $\mathbf{g}$ and S via the $\lambda\|\mathbf{g}\|^2 S$ term, encoding the “double explosion” mechanism of Theorem 11;
- A **three-phase diagram** mapping the EQUALITY, STRATIFIED, and EXPLOSIVE phases as functions of mean bias $\bar{g}$ and inequality Δ_S^2 ;
- **Nucleation theory** for honest domains, with critical radius $R_c \propto \sigma_g/\Delta f$ and Arrhenius nucleation rate;
- **Coarsening** with the classical $L(t) \sim t^{1/3}$ LSW scaling, providing testable predictions for post-reform institutional consolidation timescales;
- Explicit **theorem connections**: Thm11 $\leftrightarrow$ spinodal phase transition, Thm10 $\leftrightarrow$ domain wall pinning, Thm12 $\leftrightarrow$ defect-mediated nucleation.

The phase field formalism unifies the static geometry of previous SCX work (Theorems 10–12) with the dynamical processes (nucleation, growth, coarsening, phase transitions) that determine *how* and *when* those geometric instabilities manifest. It provides

a predictive, quantitative framework for understanding the evolution of social inequality and institutional integrity.

Final thought: In materials science, phase field theory taught us that microstructural evolution is governed by universal principles —curvature-driven motion, diffusion-limited growth, nucleation barriers —that transcend the details of specific materials. In social science, the same universality may hold. The Allen-Cahn and Cahn-Hilliard equations do not care whether the order parameter is atomic concentration or social bias. The mathematics of phase ordering is universal.

我们将 *SCX* 框架中的不平等与共识动力学表述为**相场理论**, 提供了对规范场演化 (*Allen-Cahn*)、势能场演化 (*Cahn-Hilliard*)、耦合、三相图、成核理论、粗化标度律和定理联系的统一动力学处理。相场形式化将先前 *SCX* 工作的静态几何 (定理 10–12) 与决定这些几何不稳定性如何及何时显现的动力学过程统一起来。最后思考: 相场理论教会我们, 微结构演化由超越具体材料细节的普适原理支配——曲率驱动运动、扩散限制生长、成核势垒。社会科学中同样的普适性可能成立。*Allen-Cahn* 和 *Cahn-Hilliard* 方程不关心序参量是原子浓度还是社会偏差。相有序化的数学是普适的。

A Phase Field Parameters and SCX Interpretation

B 相场参数与 SCX 解释

Parameter / 参数	Physical Meaning / 物理意义	SCX Interpretation / SCX 解释
A	Quartic coefficient in f_g	Dishonesty saturation strength / 不诚实饱和强度
B	Quadratic coefficient in f_g	Propensity for bias / 偏差倾向性
κ_g	Gauge gradient stiffness	Social cohesion / conformity pressure / 社会凝聚力
κ_S	Potential gradient stiffness	Social mobility resistance / 社会流动阻力
α	Harmonic potential strength	“Fairness” restoring force / 公平恢复力
β	Quartic saturation	Extreme inequality penalty / 极端不平等惩罚
S_0	Natural potential level	Baseline prosperity / 基线繁荣水平
λ	Gauge-potential coupling	Bias-inequality synergy (Thm11) / 偏差-不平等协同
M_g	Gauge mobility	Rate of attitude change / 态度变化速率
M_0	Base potential mobility	Base rate of resource flow / 资源流动基础速率
γ	Mobility bias	Matthew effect strength / 马太效应强度
δ_c	Domain wall thickness	Moral boundary sharpness / 道德边界锐度
σ_g	Interface energy	Social cost of difference / 差异的社会成本
R_c	Critical nucleus radius	Minimum reform coalition size / 最小改革联盟规模

Table 1: Phase field parameters and their SCX interpretations.

C Derivation of the Interface Energy

D 界面能的推导

For the 1D equilibrium domain wall with $\lambda = 0$, the Euler-Lagrange equation is obtained from the first integral:

$$\frac{\kappa_g}{2} \left(\frac{d\mathbf{g}}{dx} \right)^2 = f_g(\mathbf{g}) \quad (38)$$

This gives the kink profile. The interface energy is then:

$$\sigma_g = \int_{-\infty}^{\infty} \left[f_g(\mathbf{g}(x)) + \frac{\kappa_g}{2} \left| \frac{d\mathbf{g}}{dx} \right|^2 \right] dx \quad (39)$$

$$= \int_{-\infty}^{\infty} 2f_g(\mathbf{g}(x)) dx \quad (\text{using Eq. ??}) \quad (40)$$

$$= 2 \int_0^{g_{eq}} \sqrt{2\kappa_g f_g(\mathbf{g})} d\mathbf{g} \quad (41)$$

Evaluating with $f_g(\mathbf{g}) = \frac{A}{4} \|\mathbf{g}\|^4 - \frac{B}{2} \|\mathbf{g}\|^2$ (referenced to the minima so $f_g(g_{eq}) = 0$):

$$\sigma_g = \frac{2\sqrt{2\kappa_g}}{3} \cdot \frac{B^{3/2}}{A} \quad (42)$$

通过能量首次积分 $\frac{\kappa_g}{2} (d\mathbf{g}/dx)^2 = f_g(\mathbf{g})$ 得到扭结轮廓。界面能通过 $\sigma_g = 2 \int_0^{g_{eq}} \sqrt{2\kappa_g f_g(\mathbf{g})} d\mathbf{g}$ 计算, 得出 $\sigma_g = \frac{2\sqrt{2\kappa_g}}{3} B^{3/2}/A$ 。

E Derivation of the LSW $t^{1/3}$ Law

F LSW $t^{1/3}$ 律的推导

Following the classical LSW theory [?, ?], in the late-stage coarsening regime with a small volume fraction of the minority phase, the growth rate of a domain of radius R is:

$$\frac{dR}{dt} = \frac{D}{R} \left(\frac{1}{R_c} - \frac{1}{R} \right) \quad (43)$$

where D is the effective diffusivity and R_c is the critical radius (domains with $R = R_c$ neither grow nor shrink). Mass conservation implies $R_c = \langle R \rangle$. The asymptotic analysis yields $R_c(t) \propto t^{1/3}$ and a universal scaled size distribution.

For a system with finite volume fraction (typical for social systems), the scaling remains $t^{1/3}$ but the distribution broadens [?].

遵循经典 LSW 理论, 晚期粗化区中畴的生长率为 $dR/dt = (D/R)(1/R_c - 1/R)$ 。质量守恒给出 $R_c = \langle R \rangle$ 。渐近分析产生 $R_c(t) \propto t^{1/3}$ 。对于有限体积分数 (社会系统的典型情况), 标度保持 $t^{1/3}$, 但分布变宽。

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